Digital Logic Design

Assignment 2

Boolean Algebra & Simplification Methods | K-Map | Quine-McCluskey Method

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Question 1

Write an expression that is 1 when one or more of its variables (A, B, C, D, E) are 0.

The expression must output **1 whenever at least one variable equals 0**. When a variable equals 0, its complement equals 1. So we OR all the complements together:

X = A' + B' + C' + D' + E'

Proof using De Morgan's Law:

- X = 0 only when ALL five variables are 1, i.e. X' = A·B·C·D·E
- Applying De Morgan's: X = (ABCDE)' = A' + B' + C' + D' + E'
- This is equivalent to a 5-input NAND gate.

Verification:

- A=0, B=C=D=E=1 → X = 1+0+0+0+0 = **1** ✓
- A=B=C=D=E=1 → X = 0+0+0+0+0 = **0** ✓

Question 2

Find the value of X for all possible values of the variables: X = (A' + B')(A + B)

Step 1 — Expand using the distributive law:

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X = (A' + B')(A + B)
= A'·A + A'·B + B'·A + B'·B
= 0 + A'B + AB' + 0 [since A'A = 0 and B'B = 0]
= A'B + AB'
```

X = A'B + AB' (This is the XOR function: X = A ⊕ B)

Step 2 — Truth Table:

A	B	A'	B'	(A'+B')	(A+B)	X = (A'+B')(A+
0	0	1	1	1	0	0
0	1	1	0	1	1	1
1	0	0	1	1	1	1
1	1	0	0	0	1	0

Question 3

Write the Boolean expression for each of the logic gates in the Figure.

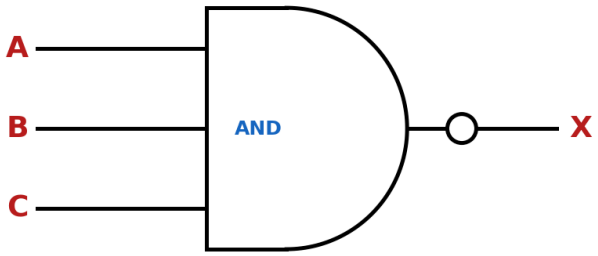
The figure shows inputs A, B, C entering an **AND gate** followed by a **bubble (inverter)** at the output — this is a **3-input NAND gate**.

X = (ABC)' = A' + B' + C' [De Morgan's Law]

Gate breakdown:

- AND section produces: ABC
- The bubble (small circle) at the output inverts the AND result
- Combined: X = (ABC)' — this is the definition of a NAND gate
- By De Morgan's theorem: (ABC)' = A' + B' + C'

Question 3: 3-Input NAND Gate → X = (ABC)' = A' + B' + C'



Question 4

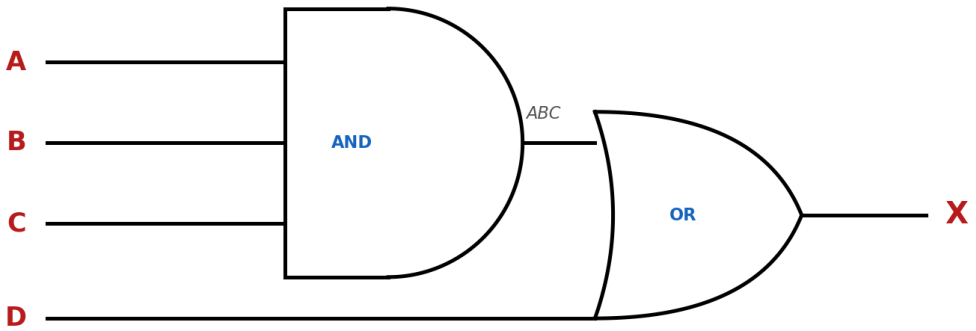
Draw the logic circuit represented by the expression: **ABC + D**

Circuit Analysis:

- **Stage 1 — AND gate (3 inputs):** Inputs A, B, C are fed into a 3-input AND gate. The output of this gate is the product term **ABC**.
- **Stage 2 — OR gate (2 inputs):** The AND gate output (ABC) and the variable **D** are fed into a 2-input OR gate. The final output is **X = ABC + D**.

X = ABC + D requires: 3-input AND gate → 2-input OR gate

Logic Circuit for: X = ABC + D



AND gate: 3 inputs (A, B, C) → produces ABC
OR gate: 2 inputs (ABC, D) → produces ABC + D

Signal flow:

- A, B, C → AND gate → ABC (intermediate signal)
- ABC, D → OR gate → X = ABC + D (final output)

Question 4 (continued) — Table 4-15: Standard SOP & POS Expressions

For each truth table, identify minterms (X=1) to build the Standard SOP, and maxterms (X=0) for the Standard POS.

Table (a) — 3 variables (A, B, C):

- X = 1 for: ABC=001(m1), 100(m4), 101(m5), 111(m7)

- X = 0 for: m0, m2, m3, m6 → maxterms

Standard SOP: X = A'B'C + AB'C' + AB'C + ABC Standard POS: X = (A+B+C)(A+B'+C)(A'+B+C')(A'+B'+C) Simplified SOP: X = AB' + A'B'C + BC

Table (b) — 3 variables (A, B, C):

- X = 1 for: ABC=101(m5), 110(m6), 111(m7)
- X = 0 for: m0, m1, m2, m3, m4 → maxterms

Standard SOP: X = AB'C + ABC' + ABC Standard POS: X = (A+B+C)(A+B+C')(A+B'+C)(A+B'+C')(A'+B+C) Simplified SOP: X = AB + AC = A(B + C)

Table (c) — 4 variables (A, B, C, D):

- X = 1 for minterms: 0000(m0), 0001(m1), 0011(m3), 0101(m5), 0110(m6), 1001(m9), 1100(m12)

Standard SOP: X = Σm(0, 1, 3, 5, 6, 9, 12)

Table (d) — 4 variables (A, B, C, D):

- X = 1 for minterms: 0010(m2), 0100(m4), 0101(m5), 0111(m7), 1001(m9), 1011(m11), 1100(m12), 1111(m15)

Standard SOP: X = Σm(2, 4, 5, 7, 9, 11, 12, 15)

Question 5

Question 5 was not specified in the assignment PDF. Please provide the question for a complete solution.

Question 6 — K-Map Simplification to Minimum SOP

Use a Karnaugh map to simplify each expression to a minimum SOP form.

Part (a): A'B'C' + AB'C + A'BC + ABC'

Step 1 — Identify minterms:

- A'B'C' = m0 (000)
- A'BC = m3 (011)
- AB'C = m5 (101)
- ABC' = m6 (110)

Step 2 — Fill the 3-variable K-map:

K-Map Q6(a): A'B'C' + AB'C + A'BC + ABC'
Simplified: No adjacent pairs → Cannot simplify: A'B'C' + A'BC + AB'C + ABC'

	BC=00	BC=01	BC=11	BC=10
A=0	1 m0	0 m1	1 m3	0 m2
A=1	0 m4	1 m5	0 m7	1 m6

Step 3 — Find adjacent groups:

- m0(000) is adjacent to: m1, m2, m4 — none are in the minterm set
- m3(011) is adjacent to: m1, m2, m7 — none are in the minterm set
- m5(101) is adjacent to: m1, m4, m7 — none are in the minterm set
- m6(110) is adjacent to: m2, m4, m7 — none are in the minterm set

This is the **even-parity (checkerboard) pattern**. No two minterms are adjacent, so no grouping is possible.

Minimum SOP: X = A'B'C' + A'BC + AB'C + ABC' (cannot be simplified further)

Part (b): DEF' + D'EF' + D'E'F'

Step 1 — Identify minterms (D = MSB, E, F = LSB):

- D'E'F' = m0 (000)
- D'EF' = m2 (010)
- DEF' = m6 (110)

Step 2 — Fill the 3-variable K-map:

K-Map Q6(b): DEF' + D'EF' + D'E'F'
Simplified: Groups: {m0,m2}=D'F', {m2,m6}=EF' → Simplified: D'F' + EF'

	BC=00	BC=01	BC=11	BC=10
A=0	1 m0	0 m1	0 m3	1 m2
A=1	0 m4	0 m5	0 m7	1 m6

Step 3 — Find adjacent groups:

- **Group 1:** {m0, m2} → differ only in E → both have D=0, F=0 → term: **D'F'**
- **Group 2:** {m2, m6} → differ only in D → both have E=1, F=0 → term: **EF'**
- All minterms covered: m0 by Group 1, m6 by Group 2, m2 shared ✓

Minimum SOP: X = D'F' + EF' = F'(D' + E)

Question 7 — List the Minterms

Expression: X = ABC + A'B'C + ABC' + AB'C + A'BC

Convert each product term to its minterm number (A=MSB, C=LSB):

Product Term	A	B	C	Binary	Minterm
ABC	1	1	1	111	m7
A'B'C	0	0	1	001	m1
ABC'	1	1	0	110	m6
AB'C	1	0	1	101	m5
A'BC	0	1	1	011	m3

Minterms: X(A, B, C) = Σm(1, 3, 5, 6, 7)

Simplification check via K-map grouping:

- {m1, m3, m5, m7}: all have C=1 → group of 4 → term: **C**
- {m6, m7}: A=1, B=1 → already covered by C; m6 alone with nothing adjacent
- m6(110) and m7(111) share AB=11 → term: **AB**
- m6 is not covered by C (since m6 has C=0), so we need the AB group

Simplified SOP: X = C + AB

Question 8 — Section 4.9, Q44: K-Map Reduction (Parts a–e)

4-variable K-map reductions. Variables: A (MSB), B, C, D (LSB). K-map uses Gray code ordering (00, 01, 11, 10) for both rows and columns.

Part (a): X = Σm(1, 3, 5, 7, 9, 11, 13, 15)

Q8(a): X = Σm(1,3,5,7,9,11,13,15)
Simplified: All D=1 cells → X = D

	CD=00	CD=01	CD=11	CD=10
AB=00	0 m0	1 m1	1 m3	0 m2
AB=01	0 m4	1 m5	1 m7	0 m6
AB=11	0 m12	1 m13	1 m15	0 m14
AB=10	0 m8	1 m9	1 m11	0 m10

Analysis:

- All 8 minterms have D = 1 (LSB = 1)
- They fill the entire D=1 column — one group of 8
- Only D is constant across all cells → term: D

Simplified: X = D

Part (b): X = Σm(0, 1, 4, 5, 8, 9, 12, 13)

Q8(b): X = Σm(0,1,4,5,8,9,12,13)
Simplified: All C=0 cells → X = C'

	CD=00	CD=01	CD=11	CD=10
AB=00	1 m0	1 m1	0 m3	0 m2
AB=01	1 m4	1 m5	0 m7	0 m6
AB=11	1 m12	1 m13	0 m15	0 m14
AB=10	1 m8	1 m9	0 m11	0 m10

Analysis:

- All 8 minterms have C = 0 (second bit from right = 0)
- They fill the entire C=0 column — one group of 8
- Only C is constant and equal to 0 → term: C'

Simplified: X = C'

Part (c): X = Σm(0, 2, 4, 6, 8, 10, 12, 14)

Q8(c): X = Σm(0,2,4,6,8,10,12,14)
Simplified: All D=0 cells → X = D'

	CD=00	CD=01	CD=11	CD=10
AB=00	1 m0	0 m1	0 m3	1 m2
AB=01	1 m4	0 m5	0 m7	1 m6
AB=11	1 m12	0 m13	0 m15	1 m14
AB=10	1 m8	0 m9	0 m11	1 m10

Analysis:

- All 8 minterms are even numbers → D = 0 (LSB = 0)
- They fill the entire D=0 column — one group of 8
- Only D is constant and equal to 0 → term: D'

Simplified: X = D'

Part (d): X = Σm(0, 1, 2, 3, 4, 5, 6, 7)

Q8(d): X = Σm(0,1,2,3,4,5,6,7)
Simplified: All A=0 cells → X = A'

	CD=00	CD=01	CD=11	CD=10
AB=00	1 m0	1 m1	1 m3	1 m2
AB=01	1 m4	1 m5	1 m7	1 m6
AB=11	0 m12	0 m13	0 m15	0 m14
AB=10	0 m8	0 m9	0 m11	0 m10

Analysis:

- All 8 minterms have A = 0 (MSB = 0)
- They fill the first two rows of the K-map — one group of 8
- Only A is constant and equal to 0 → term: A'

Simplified: X = A'

Part (e): X = Σm(0, 2, 8, 10)

Q8(e): X = Σm(0,2,8,10)
Simplified: Group {m0,m2,m8,m10}: B=0,D=0 → X = B'D'

	CD=00	CD=01	CD=11	CD=10
AB=00	1 m0	0 m1	0 m3	1 m2
AB=01	0 m4	0 m5	0 m7	0 m6
AB=11	0 m12	0 m13	0 m15	0 m14
AB=10	1 m8	0 m9	0 m11	1 m10

Analysis:

- m0=0000, m2=0010, m8=1000, m10=1010
- All four have B = 0 and D = 0
- They form a 2×2 square in the K-map (A and C vary, B and D fixed at 0)
- Two variables constant (B=0, D=0) with 4 cells → term: B'D'

Simplified: X = B'D'

Summary of All Answers

Q	Problem	Final Answer
1	Expr = 1 when ≥1 variable is 0	A' + B' + C' + D' + E'
2	X = (A'+B')(A+B)	A'B + AB' (A XOR B)
3	Boolean expr for logic gate	X = (ABC)' = A'+B'+C'
4	Circuit for ABC + D	3-input AND + 2-input OR
5	Not specified	—
6a	A'B'C' + AB'C + A'BC + ABC'	Cannot simplify (checkerboard)
6b	DEF' + D'EF' + D'E'F'	D'F' + EF'
7	List minterms of given expression	Σm(1,3,5,6,7) → simplified: C + AB
8a	Σm(1,3,5,7,9,11,13,15)	X = D
8b	Σm(0,1,4,5,8,9,12,13)	X = C'
8c	Σm(0,2,4,6,8,10,12,14)	X = D'
8d	Σm(0,1,2,3,4,5,6,7)	X = A'
8e	Σm(0,2,8,10)	X = B'D'